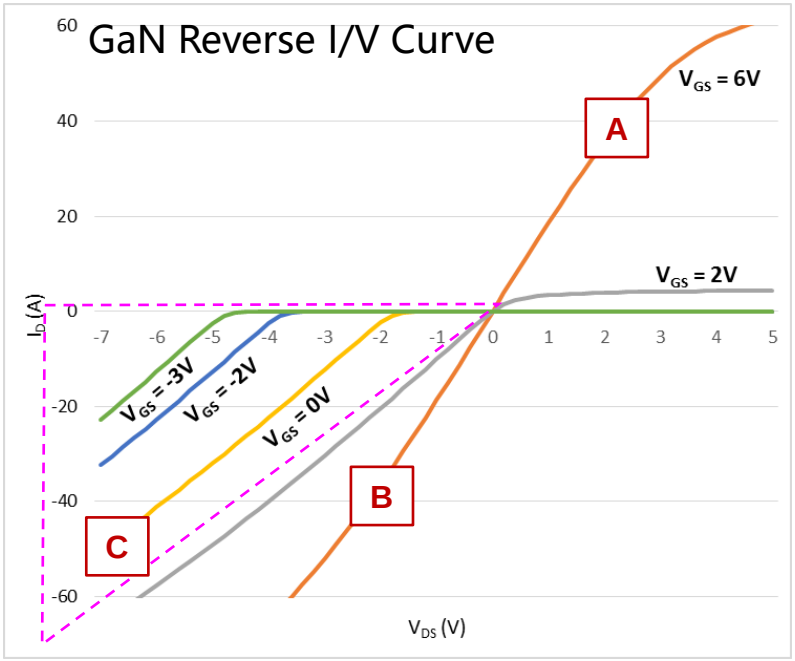
Reverse Conduction Characteristics



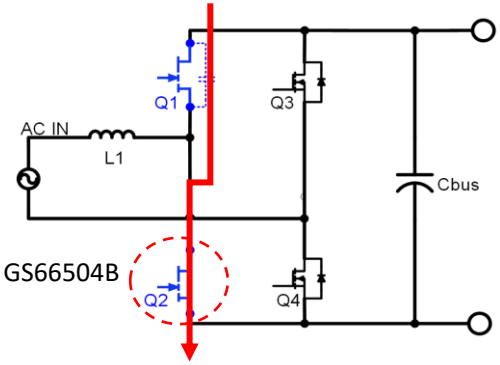
	Gate	GaN E-HEMT	MOSFET	Si IGBT
A	ON	$R_{DS(ON)}$		
B	ON	$R_{DS(ON)}$		
C	OFF	$V_{TH}+V_{GS(OFF)}$, $R_{DS(REV)}$		

- When gate is OFF (during dead time), 2DEG exhibits like a diode with $V_F = V_{TH(GD)} + V_{GS(OFF)} + I_{SD} * R_{SD(ON)}$

- There is **no body diode** (as with Si and SiC MOSFETs)
- But, GaN 2DEG can conduct in 3rd quadrant – **No need for anti-parallel diode** (as in Si IGBT)

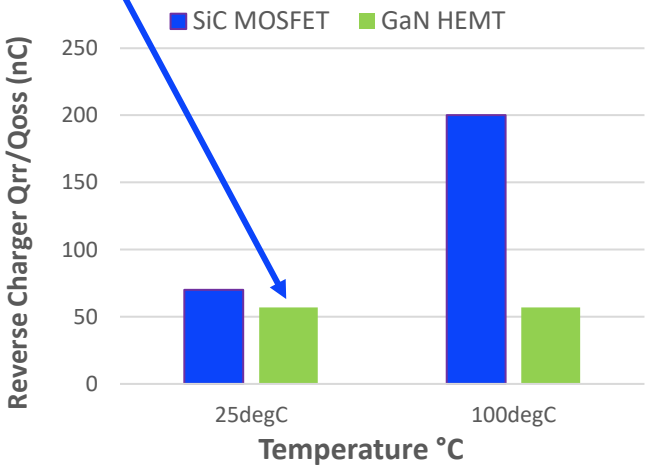
For more info: https://gansystems.com/wp-content/uploads/2020/01/Common-misconceptions-about-the-MOSFET-body-diode.pdf

Zero Reverse Recovery

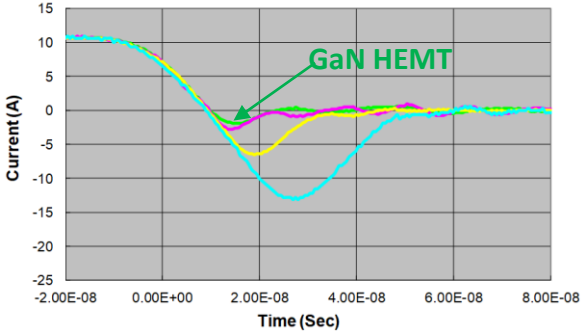




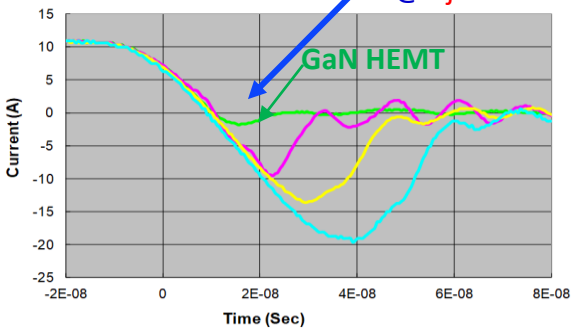
- **Qoss only,**
- **There is zero reverse recovery in GaN HEMT**



Reverse Characteristics @ Tj=25°C



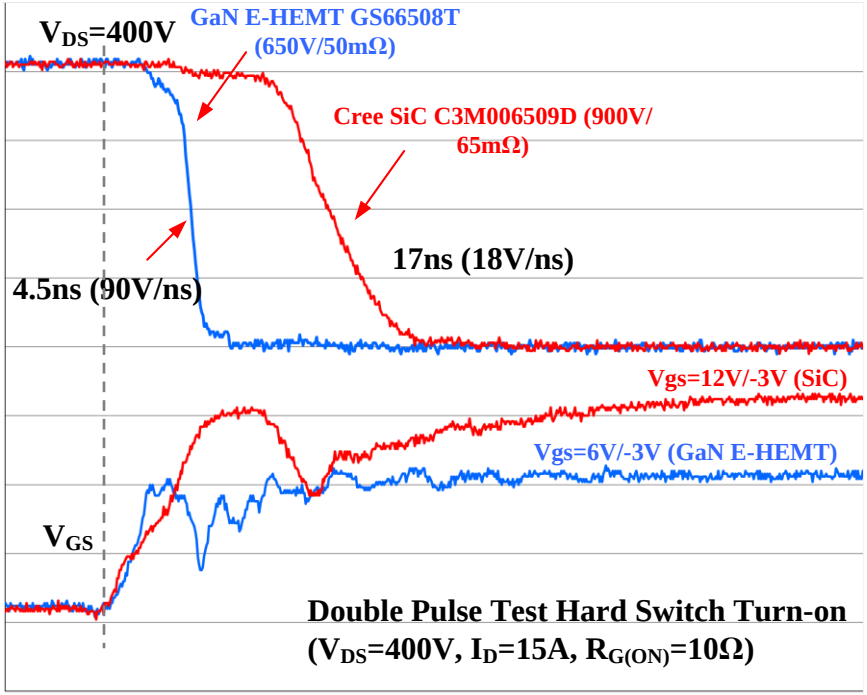
Reverse Characteristics @ Tj=100°C

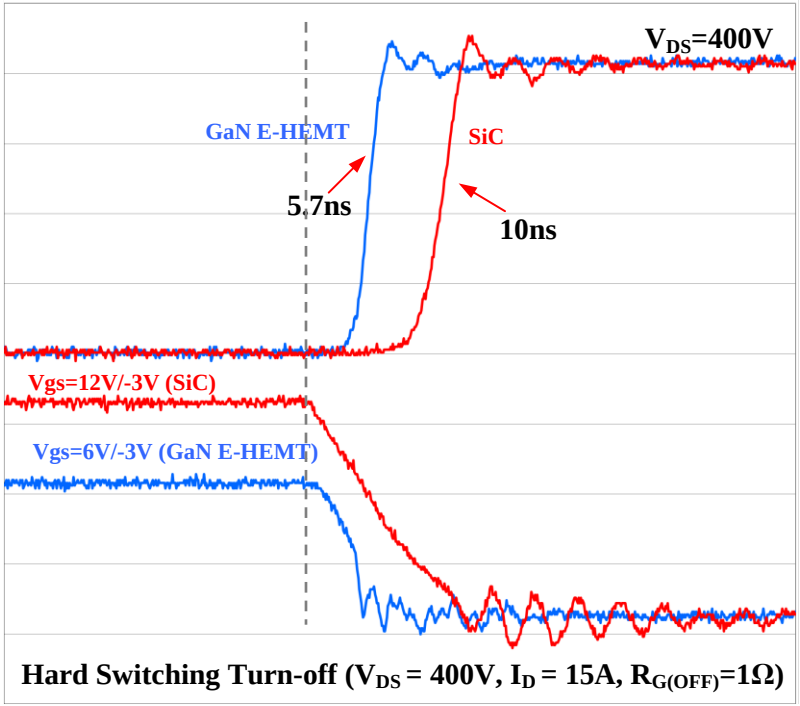


GaN HEMT 650V/15A SiC MOSFET 650V/120mΩ Si MOSFET 650V/15A Si diode 650V/15A

❖ Zero reverse recovery results in **lower switching loss** and **less EMI noise**

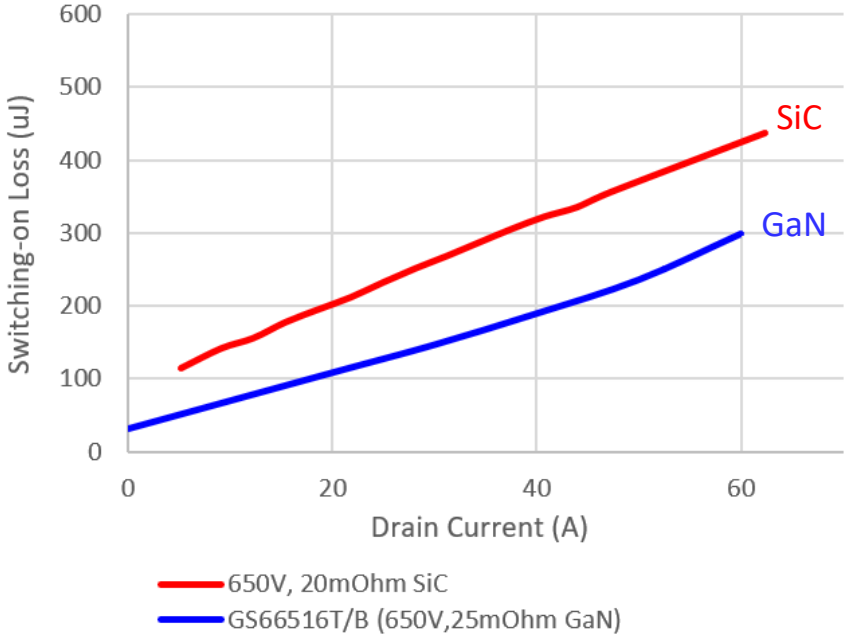
Faster Switching Transition

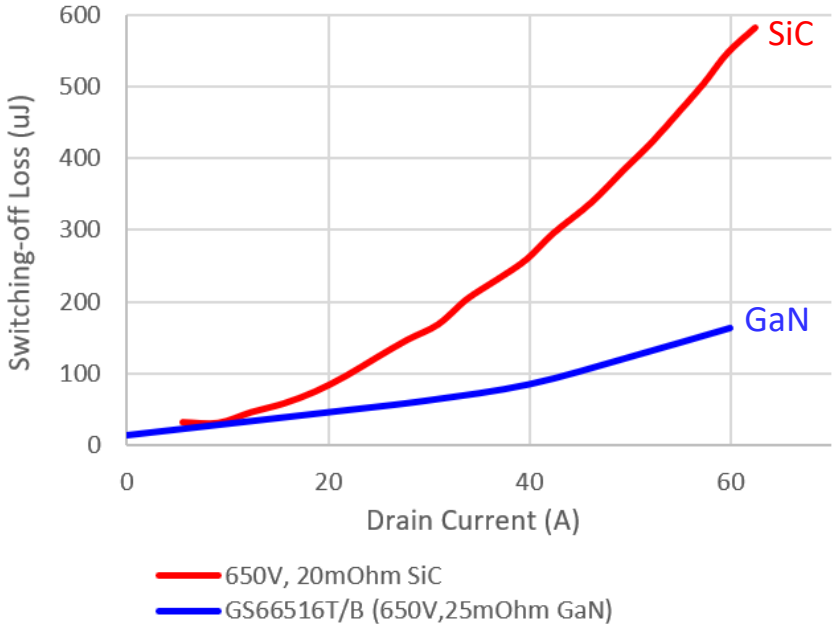




- GaN has **4x faster turn-on** and **2x faster turn-off** than state of art SiC MOSFET with similar $R_{DS(ON)}$
- Faster switching transition results in **lower switching loss**
- Layout is important for maximize the performance of GaN HEMTs. For more info: GN009 https://gansystems.com/

Switching Energy





❖ The switching loss of a GaN HEMT is **significantly lower** than 650V SiC MOSFET with similar $R_{DS(ON)}$